The Credibility Modelling and Analysis of AMI Measurements for Distribution System State Estimation

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Abstract: As fundamental equipment of advanced metering infrastructure (AMI), smart meters provide system wide visibility for distribution network. However, smart meter measurements often are not synchronized which may cause inaccuracy and divergence when directly applied to distribution system state estimation (DSSE). In order to achieve higher accuracy, the load variation between the near real-time measurements and real-time measurements which is defined as credibility are considered with the historical residential loads taken by high sample rate meters. According to the statistical result, the load variation is exponential distributed at the 95% confidence level which is utilized to update the measurements weight and 95% confidence intervals of state variables are described. A 116-node distribution network is selected to validate the proposed method. And, simulation results demonstrated its accuracy and sensitivity.

Key words—Distribution system state estimation (DSSE), nonsynchronized AMI measurements, credibility model, 95% confidence interval of system states

I. INTRODUCTION

Distribution system state estimation (DSSE) is an important part of the distribution automation system which uses the redundant measurements to estimate the state of distribution system, realizing the functions of voltage regulation, line loss reduction, network reconstruction, et al. However, the observability of distribution system is limited with the lack of real-time measurements [1]. As fundamental equipment of advanced metering infrastructure (AMI), smart meters provide wide visibility for low-voltage distribution network [2]-[3], and enhance the accuracy of measurements from SCADA system for medium-voltage distribution network [4].

Applications of AMI measurements in DSSE have been deliberated in literatures. The approaches [5]-[10] show that AMI measurements installed at 10kV feeders and distribution transformers can improve the accuracy of DSSE. Literature [11] considers about the data compatibility analysis of AMI and SCADA data synergy for DSSE. However, these studies assume that all the AMI measurements can arrive on time and the possible transmission delays are treated as measurement errors. The automated data collection system sends orders to each smart meter one by one to get real and reactive power flows, voltage magnitudes and angles at setting sampling rate. It may take 15min or even longer to collect all meter data in a distribution area [10]. Therefore, there are delays among these meter data and if they are treated as synchronized measurements and applied to the DSSE directly, the quality will be reduced and even make an unconvergent result.

To deal with these nonsynchronized AMI measurements, the literature [12] proposes a method based on the credibility of each AMI measurement using a large number of historical meter data from a distribution transformer which is proved very effective. There is still plenty of room for further research. Firstly, the historical data is sampled every 15 min and the credibility models are linearly estimated every 900 s between 0 and 15 min which may not be precise enough. Secondly, the method is an important tool for medium-voltage DSSE without consideration of AMI measurements at low-voltage distribution network which may have more unsynchronization issues limited by the communication channel capacity. Finally, the credibility of DSSE result can also be calculated with the credibility of AMI measurements and the variation of the result can be explored. In coping with these issues, this paper proposes a method of low-voltage and medium-voltage DSSE incorporating the credibility analysis and modelling of AMI measurements.

II. WLS SOLUTION FOR DSSE

The weighted least squares (WLS) approach is the most widely used method for state estimation. The linearization of the relationship between measurements and state variables are applied to simplify the nonlinear model stated as

$$z = h(x) + v \tag{1}$$

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Where $z \in R^{m\times 1}$ represents the measurements, $x \in R^{n\times 1}$ represents the state variables, defined as voltage magnitudes and angles at all buses, $h(\cdot) \in R^{m\times 1}$ represents nonlinear measurement functions, $v \in R^{m\times 1}$ represents the noise of measurements, m represents the dimension of nonlinear measurement functions, and n represents the dimension of state variables.

The objective is to minimize the sum of measurement residuals and the function can be expressed as

$$\min \boldsymbol{F}(\boldsymbol{x}) = \left[\boldsymbol{z} - \boldsymbol{h}(\boldsymbol{x})\right]^T \boldsymbol{R}^{-1} \left[\boldsymbol{z} - \boldsymbol{h}(\boldsymbol{x})\right] \quad (2)$$

$$\boldsymbol{R} = E\left\{v^T v\right\} = \begin{bmatrix} \sigma_1^2 & & & \\ & \sigma_2^2 & & \\ & & \text{L} & \\ & & & \sigma_m^2 \end{bmatrix}_{m\times m} \quad (3)$$

Where $\boldsymbol{R} \in R^{m\times m}$ represents measurement variance matrix, and σ^2 represents the square of standard deviation of the corresponding measurement noise. The minimum value of $\boldsymbol{F}(\boldsymbol{x})$ is calculated by

$$\frac{\partial \boldsymbol{F}(\boldsymbol{x})}{\partial \boldsymbol{x}} = 2\left[\boldsymbol{z} - \boldsymbol{h}(\boldsymbol{x})\right]^T \boldsymbol{R}^{-1} \frac{\partial\left[\boldsymbol{z} - \boldsymbol{h}(\boldsymbol{x})\right]}{\partial \boldsymbol{x}} = 0 \quad (4)$$

It can be rewritten by

$$\begin{aligned} \frac{\partial \boldsymbol{F}(\boldsymbol{x})}{\partial \boldsymbol{x}} &= -\left[\boldsymbol{z} - \boldsymbol{h}(\boldsymbol{x})\right]^T \boldsymbol{R}^{-1} \boldsymbol{H}(\boldsymbol{x}) = 0 \\ &= \boldsymbol{H}^T(\boldsymbol{x}) \boldsymbol{R}^{-1} \left[\boldsymbol{z} - \boldsymbol{h}(\boldsymbol{x})\right] = 0 \end{aligned} \quad (5)$$

Where $\boldsymbol{H}(\boldsymbol{x}) = \partial \boldsymbol{h}(\boldsymbol{x}) / \partial \boldsymbol{x}$. Appling Newton-Raphson algorithm, $\boldsymbol{x}$ can be updated iteratively.

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \left[\boldsymbol{H}^{\mathrm{T}}\left(\boldsymbol{x}^{(k)}\right) \boldsymbol{R}^{-1} \boldsymbol{H}\left(\boldsymbol{x}^{(k)}\right)\right]^{-1} \boldsymbol{H}^{\mathrm{T}}\left(\boldsymbol{x}^{(k)}\right) \boldsymbol{R}^{-1} \left[\boldsymbol{z} - \boldsymbol{h}\left(\boldsymbol{x}^{(k)}\right)\right] \quad (6)$$

Where k is the iteration number. When $\left|\Delta \boldsymbol{x}^{(k)} = \boldsymbol{x}^{(k+1)} - \boldsymbol{x}^{(k)}\right|$ is less than a certain value, the iteration is complete.

III. The Credibility Modelling, Analysis and Application

A. Problem Statement and Strategy

The DSSE is executed every 10~60 min considering the limitation of data acquisition and communication. As AMI measurements are updated every 15 min, the execution rate of DSSE is suggested to be the same. It is assumed that delay of AMI measurements is within 30 min. Once the DSSE is executed, and AMI measurements are not updated during this interval, the historical measurement from previous one or two intervals with the more current time stamp will be used as the near real-time data.

To verify the accuracy of the strategy, the error between the near real-time measurements and the real-time measurements should be studied. To do so, the historical data from smart meters with nonintrusive load monitoring technology which are sampled every 0.1s are used to evaluate the load variation during every 15 min. The objective is to verify if the load variation from one sampling point to another sampling point follows the characters of one kind of distribution, which is defined as the credibility for near real-time measurements. With the above information, the WLS-based state estimation algorithm can be modified and the state variables can be described as confidence intervals instead of definite values.

B. Measurements used for credibility modelling

a) Smart meter data with high sample rate

Some smart meters with high sample rate have been installed utilizing nonintrusive load monitoring technology to collect voltage, current and other information for specific electric power uses survey. With the popularization of this technology, an increasing number of smart meters will integrate with load decomposition and monitoring modules as a built-in function.

These smart meters can provide AMI measurements without delays which can be used to simulate real-time measurements and randomly delayed measurements, and the error range between these measurements will be counted for credibility modelling. Although the number of high sample rate meters is limited, the credibility model built by the method mentioned above can be applied to a certain area of power distribution systems considering that the characteristics of various load types are regularly ruled. We use historical measurements in a month from 6 meters installed at residential customers' premises for experiment.

b) Residential load data from distribution transformer

The error resulting from measurement arrival delays are affected by the short-term load variation. When the load changes smoothly, the error will decrease, as a result, the credibility will decrease and vice versa [12]. Based on this, the credibility model should be simulated for each scenario and time segment divided according to typical daily load profiles of distribution transformers.

The typical daily load profile is estimated with residential load data of distribution transformer sampled every 15 min over a month. We can identify some trends in time series and defined the time segments as follows.

TABLE I. Time Segment

Time Segment	Time of Day
Segment 1	Valley Time: 0:00-6:30
Segment 2	Peak Time: 6:30-12:00
Segment 3	Flat time: 12:00-17:30
Segment 4	Peak Time: 17:30-23:59

c) Credibility modelling

The active power is taken as an example to build the credibility model for near real-time AMI measurements.

$$\lambda_P^{i,d,t} = \frac{P_A^{i,d,t} - P_R^{i,d,t}}{P_A^{i,d,t}} \times 100\% \quad (7)$$

Where $P_R^{i,d,t}$ represents real-time active power of customer i on day d at time instant $t \in \{0:00, 0:15, \text{L}, 23:45\}$, $P_A^{i,d,t}$ represents the related near real-time active power with randomly delayed time stamp, $\lambda_P^{i,d,t}$ represents the relative load variation between $P_A^{i,d,t}$ and $P_R^{i,d,t}$ defined as credibility. Both $P_R^{i,d,t}$ and $P_A^{i,d,t}$ can be simulated using high sample-rate smart meter data according to their time stamps. It is assumed that delay time is randomly between 0 and 30 min, in other words, there will be 31 simulations of $P_A^{i,d,t}$ for each case. Group

1 means $P_A^{i,d,t}$ with no delay, and Group 31 means $P_A^{i,d,t}$ with 30 min delay. The procedure is as follows.

Step 1: λ_P is calculated for each customer every 15 min over a month with all kinds of arrival delays.

Step 2: during each time segment mentioned in chapter 3.2, λ_P with same arrival delay are classified into same group. Thus there will be 31 groups for each of 4 time segments.

Step 3: to verify if λ_P of each group follows the characters of some kinds of distribution.

According to the statistical result, the load variations of all 124 groups are exponential distributed at the 95% confidence level. The image of one exponential distribution probability density function of is shown in Fig 1.

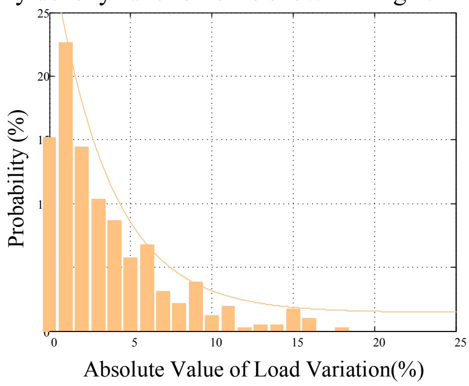

Fig. 1. The image of exponential distribution probability density function

The standard deviation of exponential distribution for each group can be seen in Fig 2.

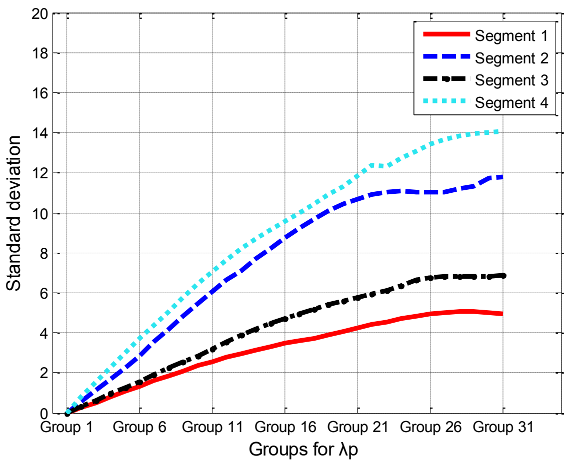

Fig. 2. The standard deviation of exponential distribution for each group

In general, the later the measurements arrive, the larger the load variation will be. Therefore, as the group number increases, the standard deviation of relative exponential distribution is growing. As we can see, the trend of parameters of the exponential distribution is reasonable at each time instant, meaning the statistical analysis result can be used as the credibility model.

C. Credibility Application

The credibility model established in chapter 3.3 can be utilized to improve the accuracy of DSSE limited by the nonsynchronized AMI measurements.

a) Measurements weights

The measurement weight is always determined only by the error of measurement equipment, causing the nonsynchronized AMI measurements have higher weights than synchronized SCADA measurements and play an important role in DSSE which will decrease the accuracy. To applicate the credibility model, the measurement weight is assumed to consist of two components, expressed as $\frac{1}{E_m+E_t}$.

Where E_m represents the error of the measurement equipment, determined as the square of accuracy of instrument. E_t represents the error of measurement arrival delay. These parameters are determined as follows.

TABLE II. TWO COMPONENTS OF MEASUREMENT WEIGHT

	E_m	E_t
SCADA measurements	4	0 （the delay of SCADA measurements is ignored）
AMI measurements	0.25	the variance of exponential distribution in chapter 3.3

b) 95% confidence intervals of state variables

It has been proved that the trend of load variations follows the character of exponential distribution shown in Fig 3.

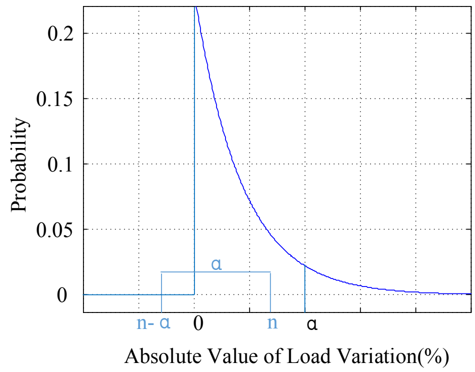

Fig. 3. Exponential distribution

When 95% is set as the confidence level, the confidence interval of the sample is $[0,\alpha]$, where $\int_0^{\alpha} 1-e^{-\frac{x}{\theta}}dx=0.95$, θ is the reciprocal of standard deviation, indicating the probability of the load variations λ_P in the group contained within $[0,\alpha]$ is expected to be 95%. According to Fig 3, the following conclusions can be drawn.

(1) $\lambda_P=\frac{P_A-P_R}{P_A}\times 100\%=0$ means the value of near real-time measurement is same with real-time measurement.

(2) The origin is between $n-\alpha$ and n, where n is the arbitrary point within the 95% confidence interval.

(3) Based on the conclusions above, the probability of P_R contained within $\left[P_A\times(1-\frac{\alpha}{100}), P_A\times(1+\frac{\alpha}{100})\right]$ is expected to be 95%.

It can be seen from the above conclusions that the measurement range for the near real-time measurement has 95% probability to contain the corresponding real-time measurement. Take the measurement range into DSSE, the range of state variables can be estimated which has 95% probability to contain the actual states. Therefore, the range of state variables is 95% confidence interval of the system states.

IV. SIMULATION

A 116-node distribution network is selected to validate the proposed method which is shown in Fig 4. The SCADA measurements are updated every 1 min, and the AMI measurements are updated every 15 min. The network states are demonstrated under the following three scenarios.

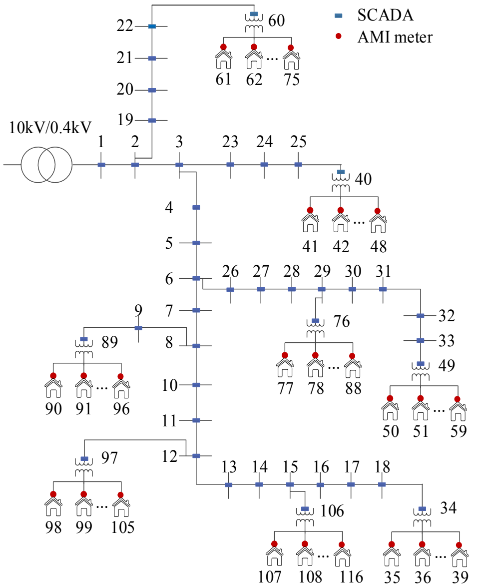

Fig. 4. 116-node distribution network

Scenario1: the measurement of each node without delay are used to estimated the distribution network states. The scenario is an ideal case and the result is just a reference for scenario 2 and scenario 3.

Scenario2: the maximum 30 min delays are randomly added to all the AMI measurements.

Scenario3: the near real-time measurements are dealt with using the proposed method.

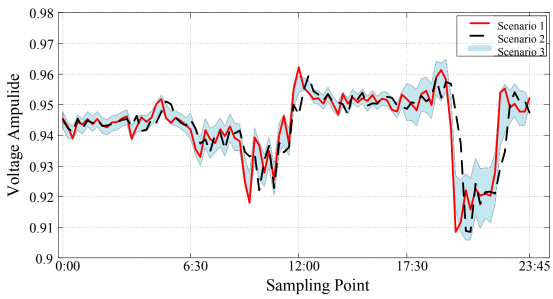

Fig. 5. State trace of voltage magnitude at node 45

The state trace of node 45 in these scenarioes are shown in Fig 5. The red profile representes the actual trace of voltage magnitude. The black profile representes the state trace estimated in Scenario 2. The blue area represents the 95% confidence interval of network state of Scenario 3. The trend of Scenario 3 is closer to the actual states, demonstrating that the result of Scenario 3 has higher accuracy and sensitivity than Scenario 2.

Average absolute relative deviation are used for accuracy assessment as well.

$$APE=\frac{1}{nb}\sum_{i=1}^{nb}\left(\frac{1}{tc}\sum_{t=1}^{tc}\left|\frac{X_{t,i}^{act,st}-X_{t,i}^{est,st}}{X_{t,i}^{act,st}}\right|\right)\times 100\% \tag{8}$$

Where $X_{t,i}^{act}$ represents the actual state of node i in time instant t, $X_{t,i}^{act}$ represents the estimated state of Scenario 2 or the average value of 95% confidence interval of Scenario 3. The state variables include voltage magnitude (VM), voltage phase angle (VA) and bus power injection (Pin). t_c represents specific time period. nb represents the number of the low voltage nodes in the system. The accuracy results are shown as follows.

TABLE III. *APE* INDEX OF 5 PERIODS

APE（%）		V_M	V_A	Pin
All day long	Scenario 2	0.482	6.208	5.819
	Average value of Scenario 3	0.305	3.864	3.571
Valley Time	Scenario 2	0.192	2.874	2.962
	Average value of Scenario 3	0.162	2.142	2.189
Peak Time	Scenario 2	0.688	8.289	7.668
	Average value of Scenario 3	0.371	4.855	4.982
Flat Time	Scenario 2	0.262	4.101	3.922
	Average value of Scenario 3	0.129	2.281	2.342
Peak Time	Scenario 2	0.825	9.965	9.582
	Average value of Scenario 3	0.592	6.330	6.862

It can be seen that the estimation results of Scenario 3 are better than Scenario 2. As we can see, the credibility model can greatly improve the DSSE accuracy limited by the nonsynchronized AMI measurements.

V. CONCLUSION

Based on the results shown in this paper, the proposed method can be a modified approach to deal with the nonsynchronized measurements, and the credibility models are built in advance and the computational cost are not important comparing to the process of WLS-based state estimation. Considering that the error range of the nonsynchronized measurement is a type of interval estimate in statistics, the network states are described as confidence intervals with given confidence level, which can make the result more reasonable and more acceptable to the actual situation.

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